KAIZEN MATHS

Lesson starter

Taylor and Maclaurin Series

Maclaurin Basics

Name

S1

Write each standard Maclaurin expansion up to the requested term.

Write the Maclaurin series for cos x.

S2

Use the Maclaurin formula to form each polynomial from the derivative values.

Given f(0) = 2, f'(0) = -3, f''(0) = 8 and f'''(0) = 12, find the Maclaurin polynomial up to x^3.

S3

Expand each function as a Maclaurin series up to the requested power of x.

Expand e^2x as a Maclaurin series up to and including the term in x^4.

S4

State the interval or condition of validity for each expansion.

State the condition of validity for the Maclaurin expansion of ln(1 + 3x).

Show your working

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Show your working

KAIZEN MATHS

Exit ticket

Taylor and Maclaurin Series

Maclaurin Basics

Name _____

EA1

Write each standard Maclaurin expansion up to the requested term.

Write the Maclaurin series for sin x.

EA2

Use the Maclaurin formula to form each polynomial from the derivative values.

Given f(0) = 3, f'(0) = -2, f''(0) = -8 and f'''(0) = 30, find the Maclaurin polynomial up to x^3.

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Exit ticket

Taylor and Maclaurin Series

Maclaurin Basics

Name _____

EB1

Expand each function as a Maclaurin series up to the requested power of x.

Expand sin(2x) as a Maclaurin series up to and including the term in x^3.

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Exit ticket

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Name _____

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Write each standard Maclaurin expansion up to the requested term.

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Given f(0) = 3, f'(0) = -2, f''(0) = -8 and f'''(0) = 30, find the Maclaurin polynomial up to x^3.

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EB1

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Expand sin(2x) as a Maclaurin series up to and including the term in x^3.

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Teacher answers & working

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S1

Write each standard Maclaurin expansion up to the requested term.

Write the Maclaurin series for cos x.

Answer:

$1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots$

Working:

Use the standard Maclaurin series for cos x.

Only even powers appear.

The signs alternate: +, -, +, - .

Answer: $1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots$

S2

Use the Maclaurin formula to form each polynomial from the derivative values.

Given f(0) = 2, f'(0) = -3, f''(0) = 8 and f'''(0) = 12, find the Maclaurin polynomial up to x^3.

Answer:

$2 - 3x + 4x^2 + 2x^3$

Working:

Use $f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!}$.

Substitute the derivative values: $f(x) = 2 + -3x + \frac{8x^2}{2} + \frac{12x^3}{6}$.

Simplify each coefficient.

$f(x) = 2 - 3x + 4x^2 + 2x^3$.

S3

Expand each function as a Maclaurin series up to the requested power of x.

Expand e^2x as a Maclaurin series up to and including the term in x^4.

Answer:

$1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + O(x^5)$

Working:

Start with the standard expansion: $e^u = 1 + u + \frac{(u)^2}{2!} + \frac{(u)^3}{3!} + \frac{(u)^4}{4!} + \ldots.$

Substitute u = 2x.

After substitution: $e^{2x} = 1 + 2x + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \frac{(2x)^4}{4!} + \ldots.$

Simplify each substituted power and keep terms only up to x^4.

$e^{2x} = 1 + 2x + 2x^2 + \frac{4}{3}x^3 + \frac{2}{3}x^4 + O(x^5)$.

Teacher answers & working

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S4

State the interval or condition of validity for each expansion.

State the condition of validity for the Maclaurin expansion of ln(1 + 3x).

Answer:

$-\frac{1}{3} < x \leq \frac{1}{3}$

Working:

For ln(1 + u), the Maclaurin series is valid for -1 < u ≤ 1.

Here u = 3x.

So $-1 < 3x \leq 1$.

Divide by 3: $-\frac{1}{3} < x \leq \frac{1}{3}$.

EA1

Write each standard Maclaurin expansion up to the requested term.

Write the Maclaurin series for sin x.

Answer:

$x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots$

Working:

Use the standard Maclaurin series for sin x.

Only odd powers appear.

The signs alternate: +, -, +, - .

Answer: $x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots$

EA2

Use the Maclaurin formula to form each polynomial from the derivative values.

Given f(0) = 3, f'(0) = -2, f''(0) = -8 and f'''(0) = 30, find the Maclaurin polynomial up to x^3.

Answer:

$3 - 2x - 4x^2 + 5x^3$

Working:

Use $f(x) = f(0) + f'(0)x + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!}$.

Substitute the derivative values: $f(x) = 3 + -2x + \frac{-8x^2}{2} + \frac{30x^3}{6}$.

Simplify each coefficient.

$f(x) = 3 - 2x - 4x^2 + 5x^3$.

Teacher answers & working

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EB1

Expand each function as a Maclaurin series up to the requested power of x.

Expand sin(2x) as a Maclaurin series up to and including the term in x^3.

Answer:

$2x - \frac{4}{3}x^3 + O(x^4)$

Working:

Start with the standard expansion: sin u = u - $\frac{(u)^3}{3!}$ + ...

Substitute u = 2x.

After substitution: sin(2x) = 2x - $\frac{(2x)^3}{3!}$ + ...

Simplify each substituted power and keep terms only up to x^3.

$\sin(2x) = 2x - \frac{4}{3}x^3 + O(x^4)$.